

Manufacturing Execution System

Capability & Operations Handbook

What the resin pouring station's production system does today, screen by screen, with a working guide for each role that uses it — and an honest account of what is still experimental.

What's in this handbook

Sections 1 to 8 describe what the system does. Sections 9 to 12 are working guides for each role, written so a new starter can follow them on their first shift. Section 14 covers what exists but is not in service.

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ABOUT THE SCREENSHOTS

Every screen shown is a live capture of the running system against a representative fortnight of activity. The operator and manager names in them are fictional.

What the system is

The resin pouring station previously ran on a multi-sheet Excel tracker. It held OEE metrics, shift handover summaries and a live pace tracker, and it worked — but it had hardcoded row ranges, operator names spelled differently between sheets, downtime that was counted twice, and it could only be read by whoever had the file open.

What replaced it is a web application, opened in a browser at each station, in the office and on the floor display, backed by a PostgreSQL database. Managers dispatch work orders; operators log against them from the pump. The figures on the wall display, in the analytics charts and in the emailed shift handover come from the same rows.

18

screens, from the operator terminal to the IT console

3

roles, each seeing only what its job needs

18

database tables of production, quality and configuration

3

shifts modelled, each with its own targets and pace

24

interface themes, for readability in different light

75

automated checks run before a release

What it records

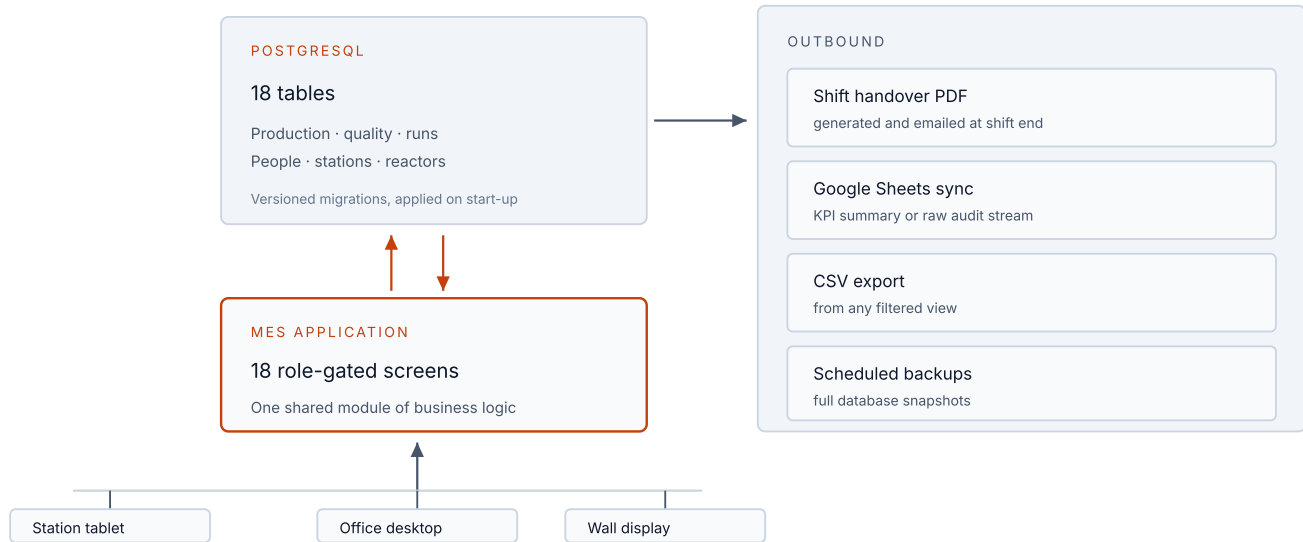
- Every unit poured, attached to an operator, a station, a resin, a lot number and a timestamp.
- A cartridge lot check against each pouring log, including the ones that failed.
- Downtime by reason and duration, and scrap split between empty and filled containers.
- Start-of-shift, changeover and spill photo audits, filed against the station.
- Pre-shift validation per operator, per shift, per pump station.
- Reactor volume, reconciled against what has been logged rather than overwritten.

SCOPE

The system is in service for resin pouring. Pack-out logging and machine integration are built but not in use; both are described in section 14 rather than here, because neither has been committed to.

One database, many windows

Nothing on the floor keeps its own copy of the numbers.



Schema changes

Structural changes ship as numbered migrations that apply themselves on start-up, rather than SQL run by hand. Logs carry foreign keys to the operator and station as well as the legacy name text, so renaming someone does not split their history.

Where the logic lives

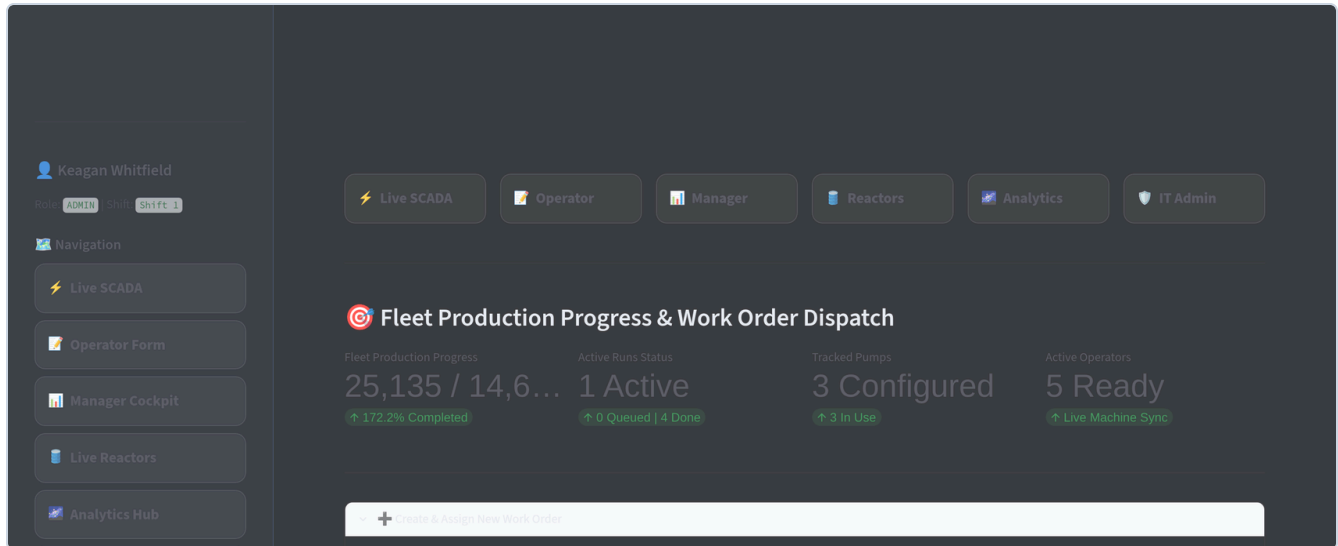
Business logic sits in one module the whole application shares. Every screen that shows a figure reads it through the same functions, so a change to how yield is calculated applies everywhere at once.

What happens when an operator submits a log

- 1 Checked at the station**
 The cartridge lot is compared against the run before the form accepts the entry.
- 2 One transaction**
 The log and its verification record are written together; neither can exist without the other.
- 3 The run re-counts**
 Progress is recomputed from the logs matching its station, resin, format and lot.
- 4 Everything else follows**
 Display, analytics, handover and exports all read those same rows.

Work orders

A manager creates a work order against a reactor, resin, container format and lot. Hourly logs submitted at that station are matched to it automatically.



Work Orders & Assigned Runs. Fleet-wide progress across active runs, with status, assigned operator, pump station and lot. Completed orders move to a separate history tab.

DISPATCH

Create and assign

Reactor, size, resin, container format, target units, station and lot number, dispatched to a station rather than to one named person.

COUNTING

Progress from logs

A run's count is recomputed from the production logs that match it, so deleting a bad entry re-syncs the order instead of leaving it overstated.

SHARED STATIONS

Several operators, one run

Progress follows the pump station, so any number of operators can work the same run without a duplicate assignment being created for each.

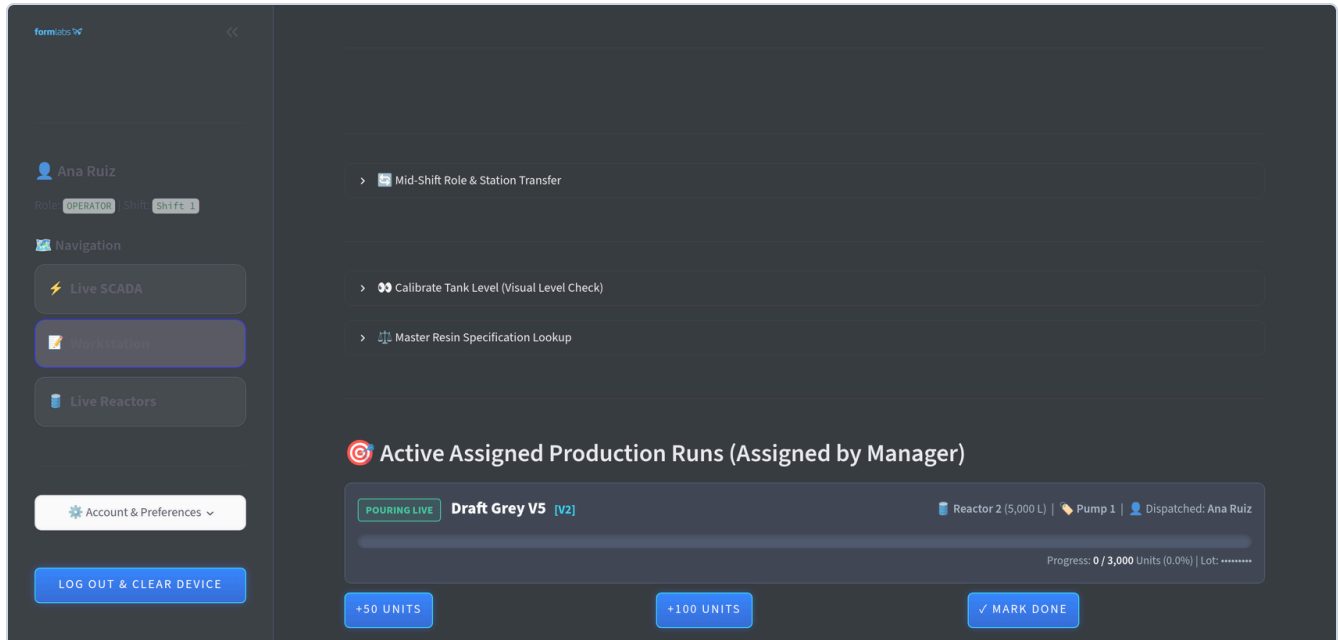
CLOSEOUT

Finish on the real number

A run can be completed on a custom final total when the physical count differs, and the reactor it drew from is released.

The operator terminal

One screen, sized for a gloved hand on a tablet: stacked inputs, large targets, no navigation to get lost in.



Active runs at this station. Reactor, container format, dispatching manager and live progress. The lot is masked for operators — see section 4.

TAB 1

Log hourly pouring

Station, container format, resin, verified lot, good units and both scrap categories, with the target fill weight and tolerance for the selected resin shown alongside.

TAB 2

Log station downtime

Reason from the plant's configurable list, duration and notes. Feeds the downtime ranking used in analytics.

TAB 3

Cleanliness & photo audit

Start of shift, end of shift, line transfer or spill, with a camera capture or upload, filed against the station.

TAB 4

Manager comms

A direct message thread with the plant lead, inside the terminal.

ALWAYS VISIBLE

Units poured, scrap and quality yield for that operator sit at the top of the screen and update as they log. It is the same figure management sees; there is no separate scoring.

Cartridge lot verification

Lot labels get mixed up at the pumps: a pallet of pre-labelled empties is staged that belongs to a different lot, and resin goes into it. Previously the logging form auto-filled the run's lot and displayed it, so an hour could be logged without a cartridge ever being turned over.

2. Cartridge Lot Verification

Expected lot for this run: hidden on purpose — read the cartridge, not the screen

Turn the cartridge over. The bottom is stamped followed by the lot. Type it exactly as printed — spacing, case and the prefix don't matter.

L — lot stamped on the cartridge bottom

3. Production Output

The check. The lot the run expects is masked. The operator reads the **L-** code stamped on the cartridge base and types it. Spacing, case and the prefix are normalised, so the stamp typed verbatim matches a run lot entered bare.

L-240880119

STOP — DO NOT POUR. This cartridge is not from the lot assigned to your run. Set it aside and get your lead.

Pulled the cartridge instead? Log the catch — it belongs in the record.

WRONG CARTRIDGE — PULLED IT, NOTHING POURED

Logging it anyway? Say what happened.

Choose an option

Details (required)

Photograph the stamp — required to log a pour against a flag.

When it doesn't match. Submission is refused until the cartridge is pulled — recorded as a catch, with no production logged — or a cause, details and a photograph of the stamp are supplied. The log then saves flagged, carrying the lot physically in the cartridge.

FREQUENCY

Full check, then a confirm

The full check runs on the first log of a shift, on any change of run, lot, resin or station, after four hours, and on every tenth log. Otherwise a single confirmation tap.

COST

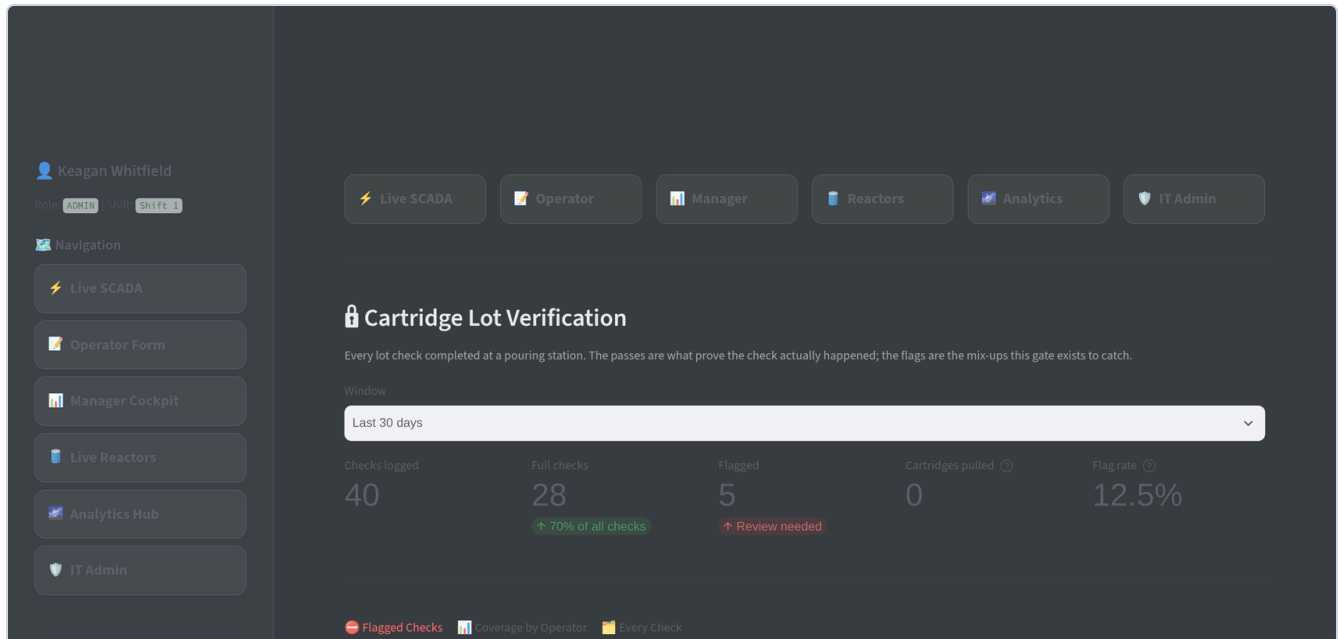
Photo only on a mismatch

A clean check is one field and one tap. The photograph is required only when logging against a flag — and deliberately not required to pull a cartridge, so the safe action is never the slower one.

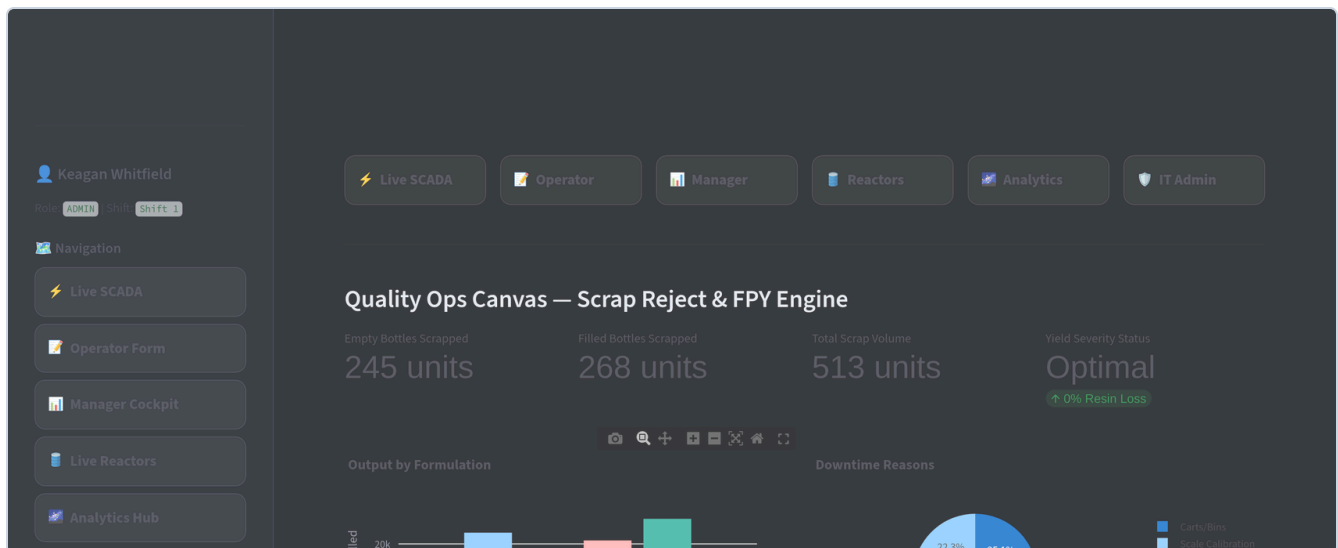
DESIGN NOTE

A mismatch does not stop the line. It requires an explanation, because a control that halts production is one people learn to work around. Every flagged pour and every pulled cartridge reaches the manager's review page.

Review and scrap intelligence



Lot Verification review. Every check completed, pass or fail. The passes record that the check happened; the flags are the mix-ups. Coverage per operator shows who is completing full checks and who is using the confirmation tap.



Scrap & Yield Intelligence. First-pass yield against the configured target, with scrap split between empty and filled containers — a handling problem and a lost-resin problem respectively — and downtime reasons ranked alongside.

PHOTO AUDITS

Cleanliness gallery

Start of shift, end of shift, line transfer and spill audits with photograph, station, operator and timestamp. Spills are counted separately.

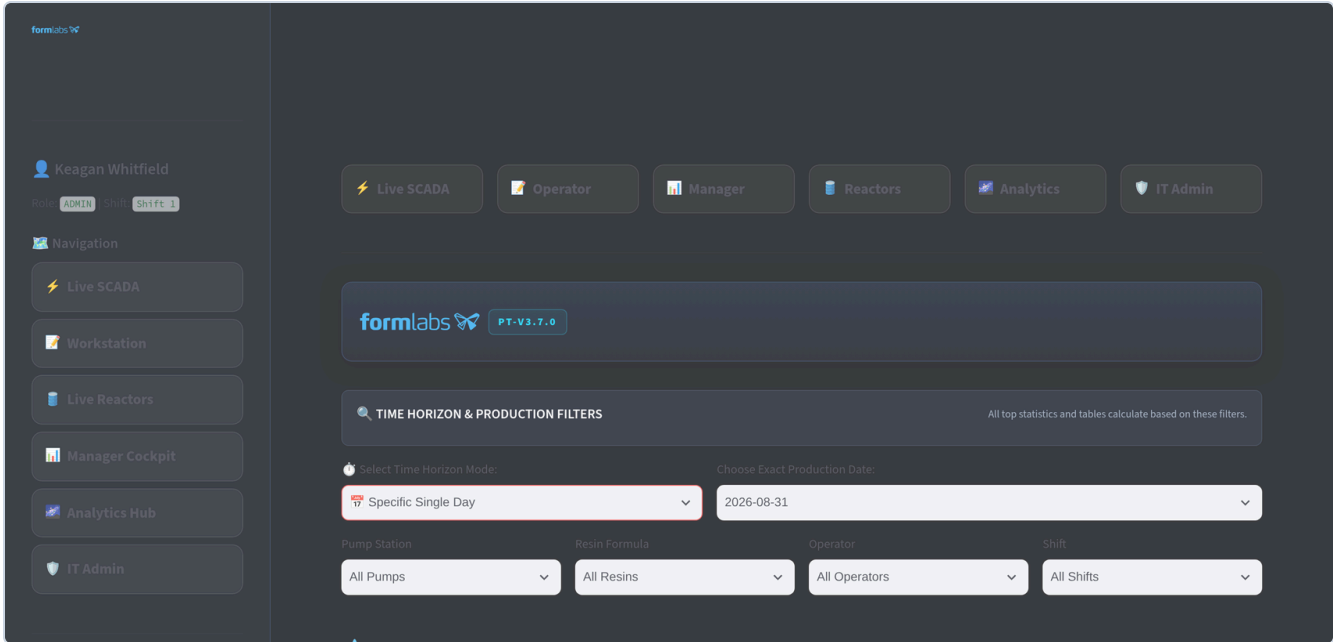
EXPORT

CSV from the register

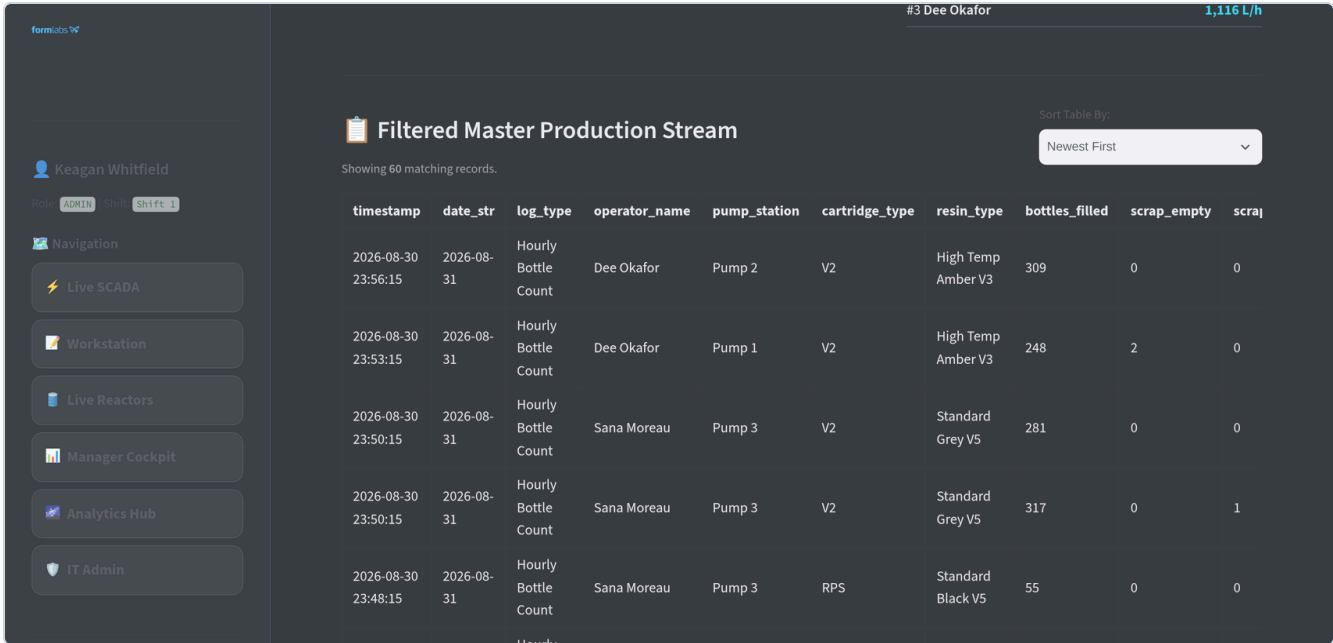
The verification register exports directly, so a quality question or customer audit can be answered without a database query.

The floor as it is now

The landing screen for every role. Filters across time horizon, station, resin, operator and shift recalculate every figure on the page.

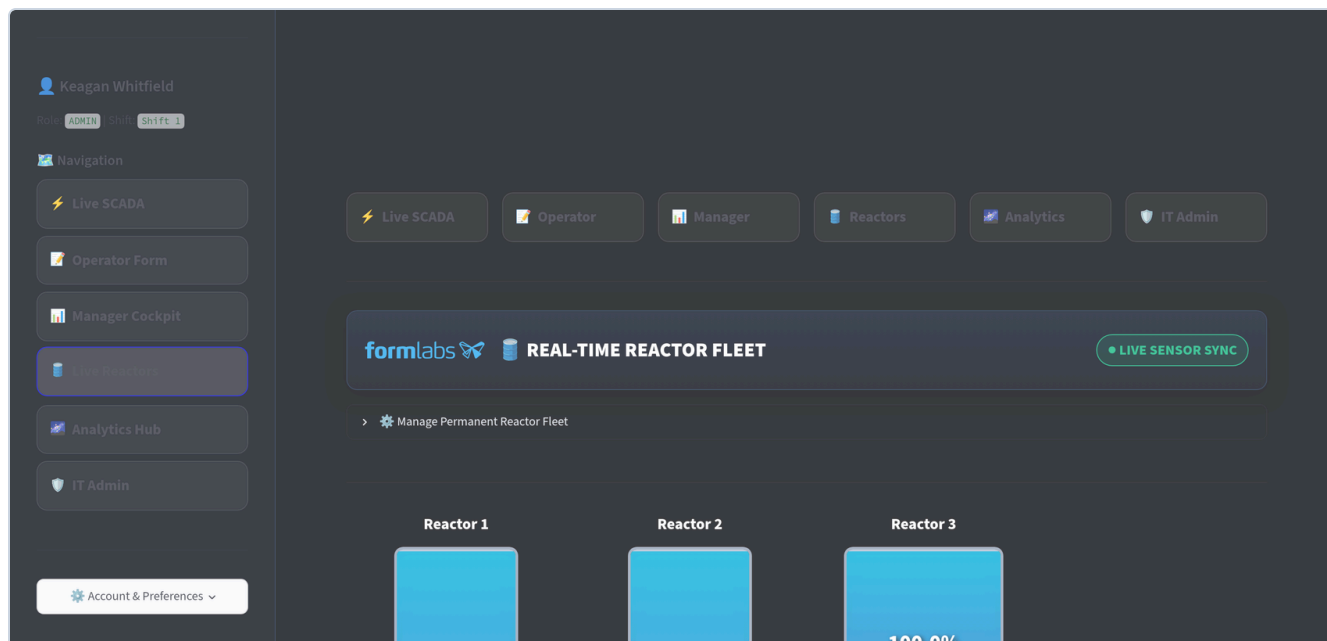


Live SCADA. Volume output, resin mass poured (calculated from each formulation's master fill weight), run velocity against the configured target, and pouring yield.

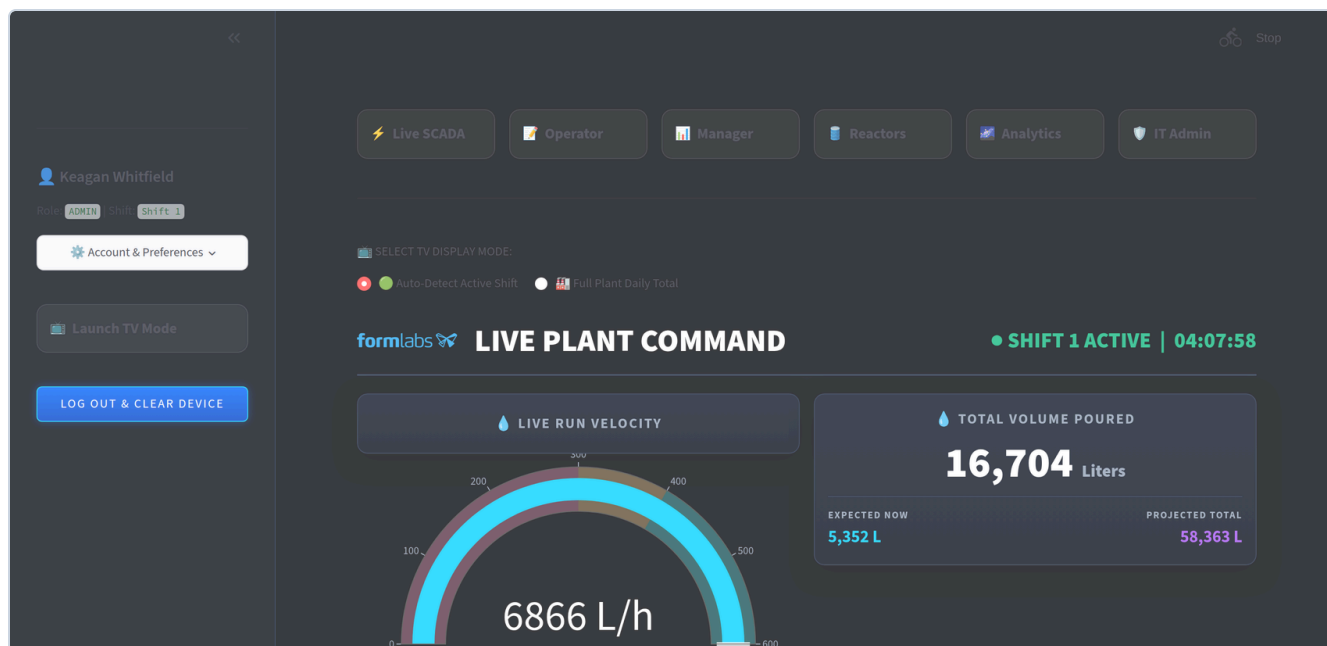


Below the fold. Shift trajectory against pace, a pouring leaderboard, and the filtered production stream — the individual logs behind the figures above.

Reactors and the floor display

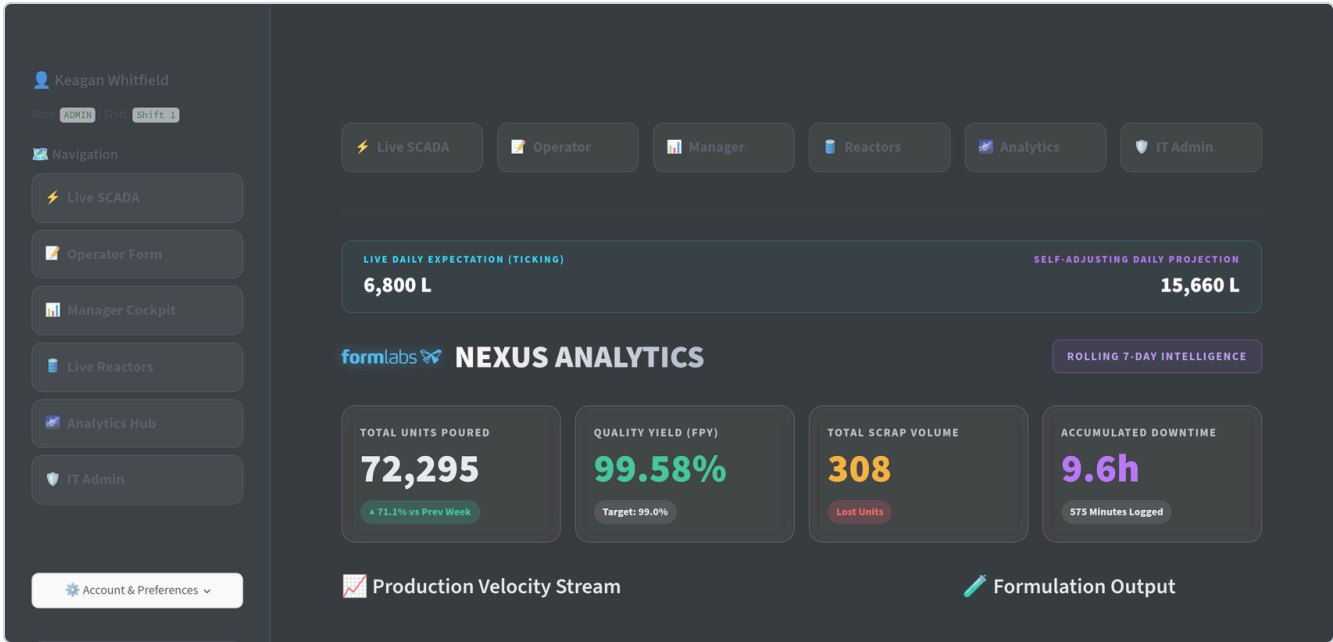


Live Reactor Fleet. Each vessel's current resin, assigned pump and remaining volume, with reconciliation by visual fill percentage or measured litres when the calculated figure has drifted from what is in the tank.

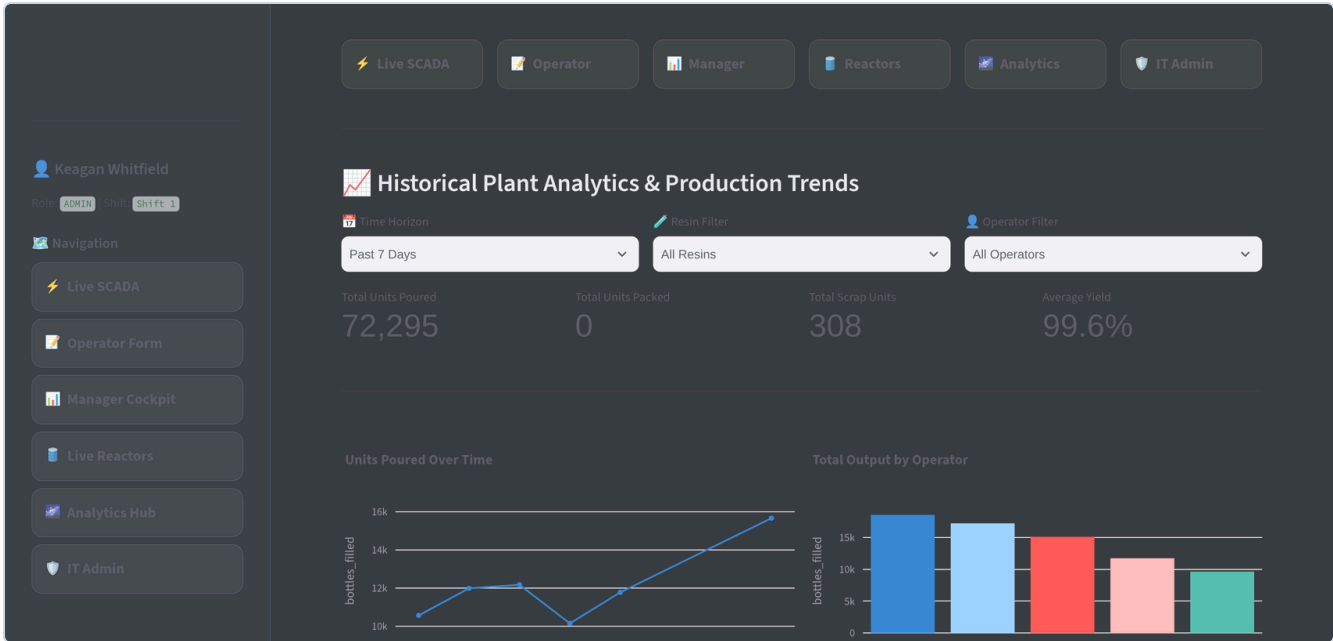


Floor display mode. A full-screen, self-refreshing view that auto-detects the active shift or shows the full plant day, so the team can see its own pace without opening anything.

Trends

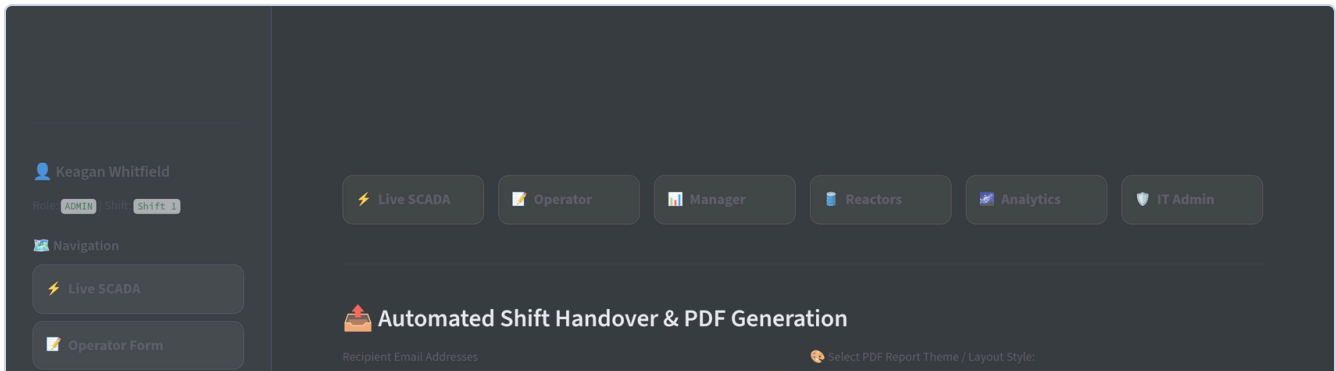


Analytics Hub. Output trend over the selected horizon, share of production by resin formulation, and downtime ranked by reason.



Historical Production Trends. The same data filtered by horizon, resin and individual operator, for month-to-date, year-to-date or any range.

Reporting out



Automated Shift Handover. Output, yield, downtime and scrap for the shift, rendered as a PDF in one of two layouts and emailed to the plant's distribution list.

CLOUD EXPORT

Google Sheets sync

Two payload shapes — an aggregated KPI summary or the raw production audit stream — across four time horizons, pushed on demand, at shift handover, or on a scheduled webhook. Columns are selectable, and internal-only fields are stripped before anything leaves the app.

SELF-SERVICE

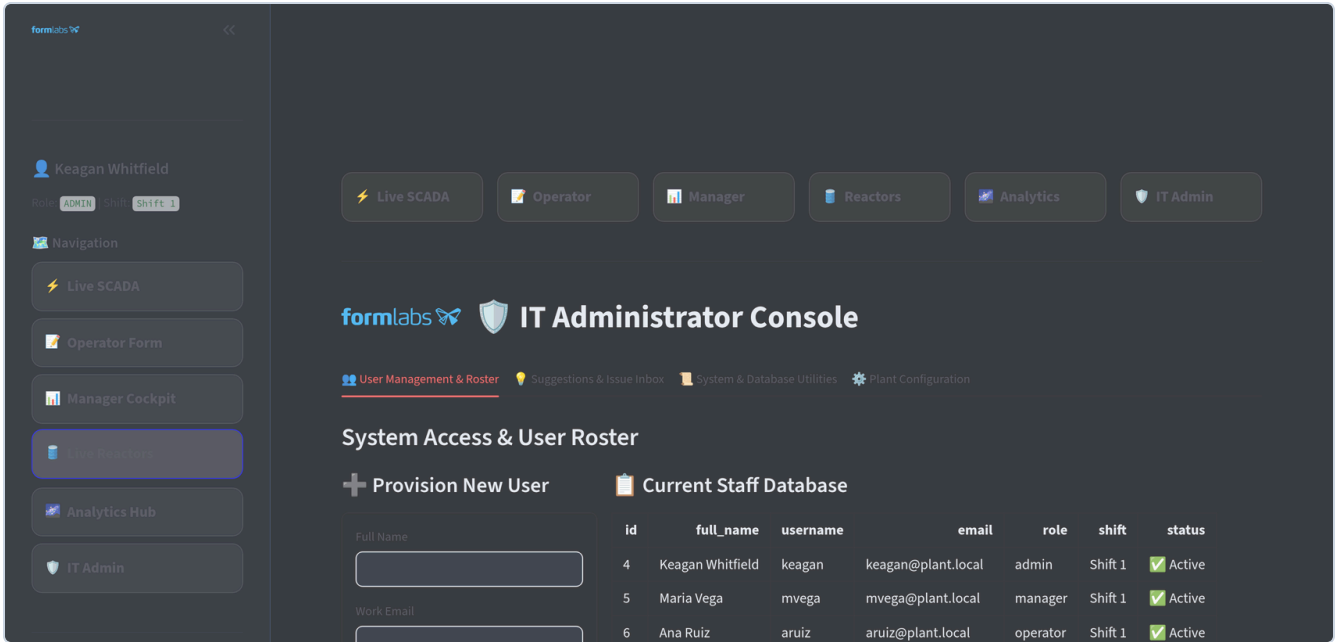
CSV from any view

Filtered log views and the verification register export directly, so finance, quality or an audit can be answered without a developer.

ONE SOURCE

The handover PDF, the Google Sheet, the CSV export and the floor display all render from the same production log rows. There is no second system to keep in step.

Running the system



IT Administration. User management and roster, the suggestions and issue inbox, database utilities, and global plant configuration.

PEOPLE

Accounts and roles

Provision staff, assign role and shift, set per-operator rate targets, reset PINs and unlock accounts. Managers can provision floor personnel without full IT access.

PLANT

Global parameters

Shift start times, lengths and break minutes for all three shifts; rate targets; yield targets; and the handover email recipients.

EQUIPMENT

Stations, reactors, reasons

Pump stations, the reactor fleet and capacities, and the downtime reason list, so the vocabulary stays the plant's own.

RECOVERY

Backup and restore

Full database snapshots on demand or on schedule, and a restore path from any snapshot.



Master Resin Specifications. SKU, internal resin code, target fill weight with tolerance band and shelf life — what lets a bottle count
convert into resin mass poured

Access control and data integrity

Accounts

- **PINs are hashed.** Salted with bcrypt; the database holds no readable PIN, including for administrators.
- **Lockout.** Five failed attempts locks the account for fifteen minutes, with an instant IT unlock.
- **Sessions are tokens.** A signed token with a fixed lifetime backs "remember this device", revocable server-side.
- **Every page checks the role.** Enforced on the page itself, so typing a URL for a screen your role lacks returns access denied.
- **Output is escaped.** Values typed on the floor are escaped before rendering, so a note cannot inject markup or break its own card.

The record

- **History survives a rename.** Logs carry foreign keys to the person and station as well as the name text.
- **Deletions re-sync.** Removing a log recalculates whatever work order it counted toward.
- **Bulk deletion is deliberately hard.** Disabled until the word DELETE is typed into a confirmation box.
- **Corrections are visible.** A plant-level correction is written as a labelled system adjustment, not by editing an operator's entries.
- **Secrets are out of the repository.** Connection strings and credentials live in environment config, not in source.

The screenshot shows the 'Log Management & Data Cleanup' interface. On the left is a sidebar with user information for Keagan Whitfield (Role: ADMIN, Shift: Shift 1) and a navigation menu with buttons for Live SCADA, Operator Form, Manager Cockpit, Live Reactors, Analytics Hub, and IT Admin. The main area has a top bar with buttons for Live SCADA, Operator, Manager, Reactors, Analytics, and IT Admin. Below this is the 'Log Management & Data Cleanup' section, which includes a warning message: 'Deletions here are permanent and cannot be undone. Deleting a production log automatically re-syncs the progress of whatever Work Order it was counted against, so totals stay accurate after cleanup.' There are two tabs: 'Production Logs (Pouring / Packing)' (active) and 'Downtime Logs'. Below the tabs are filters for 'From Date' (2026/08/18), 'To Date' (2026/08/31), 'Log Type' (All Types), 'Pump Station' (All Pumps), and 'Operator' (All Operators). At the bottom, it states '492 record(s) match the current filters.'

Log Management & Cleanup. Filter production and downtime logs by date, station, resin or operator, review, and delete with the work-order re-sync handled automatically.

Roles

Everyone signs into the same application and sees a different one. Access is checked on every page, so the difference is enforced rather than cosmetic.

CAPABILITY	OPERATOR	MANAGER	ADMIN
Live SCADA dashboard	Yes	Yes	Yes
Startup checklist & terminal unlock	Required	n/a	n/a
Log hourly pouring & lot verification	Yes	Test mode	Test mode
Log downtime & cleanliness audits	Yes	Yes	Yes
Message the plant lead	Yes	Both ways	Both ways
Live reactor fleet & reconciliation	Yes	Yes	Yes
See the lot a run expects	Masked	Yes	Yes
Create & dispatch work orders	—	Yes	Yes
Analytics hub & historical trends	—	Yes	Yes
Lot verification review & scrap intelligence	—	Yes	Yes
Shift handover PDF & Google sync	—	Yes	Yes
Floor roster & PIN resets	—	Floor staff	All users
Master resin specifications	—	Yes	Yes
Log management & bulk cleanup	—	Yes	Yes
Plant configuration & shift parameters	—	—	Yes
Database backup & restore	—	—	Yes

Test mode lets a manager or administrator log on behalf of an operator for training and troubleshooting. The entry is attributed to the operator selected and is visible in the log.

Your shift, start to finish

Operator Pours resin at a pump station	Manager Dispatches and reviews	Admin Runs the system
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1 Sign in at the station tablet

Your operator ID and PIN. Tick *Remember this device* on a tablet that stays at your station. Five wrong PINs locks the account for fifteen minutes — ask your lead for a reset rather than guessing.

2 Clear the startup checklist for your pump

The terminal stays locked until you do. Pick the pump you are starting at, submit the start-of-shift cleanliness photo for that station, then confirm the QC checksheet and that your empties and receiving carts are staged. **Moved to a different pump later? The checklist comes back for that pump** — it certifies the station you are standing at.

Terminal locked. Choose your pump here; the answer carries through to your logging tab, so you pick your station once.

3 Check the run you have been assigned

Runs assigned to your station appear at the top with reactor, container format and live progress. If nothing is listed, tell your lead before pouring — your units will still be recorded, but they will not count toward a work order.

IF YOU ARE LOCKED OUT

Five attempts, fifteen minutes

The lock releases on its own, but a manager can reset your PIN from the floor roster immediately.

SHARED TABLET

Log out at the end of your shift

Logs are attached to whoever is signed in, so the next operator's units should not land under your name.

Logging an hour of pouring

4 Set your station and material

Pump station, container format (V2, V1, RPS bulk jug or Pigment) and resin formulation. The target fill weight and tolerance for that resin appear underneath — check it against your scale.

5 Verify the cartridge lot

Turn a cartridge over and type the **L-** code stamped on its base. The lot the run expects is hidden on purpose — **read the cartridge, not the screen**. Type it as you see it; spacing, case and the prefix do not matter.

Log Hourly Pouring ⚠️ Log Station Downtime 📷 Cleanliness & Photo Audit 💬 Manager Comms

1. Station & Material Setup

Pump Station: Pump 1

Container Format: V2 (1L Cartridge)

Resin Formulation: Draft Grey V5

Target: 1000.0g | Range: 990-1010g (no spec on file for V2 (1L Cartridge) — showing this resin's spec from another Container Format)

2. Cartridge Lot Verification

Expected lot for this run: L-2503C0088 hidden on purpose — read the cartridge, not the screen

Turn the cartridge over. The bottom is stamped L- followed by the lot. Type it exactly as printed — spacing, case and the prefix don't matter.

L — lot stamped on the cartridge bottom

L-2503C0088

LOG OUT & CLEAR DEVICE

A clean check. Green means the cartridge belongs to this run. On the next log at the same station with nothing changed, this becomes a single confirmation tap — returning to the full check on any change of run, lot, resin or station, after four hours, and on every tenth log.

6 If it does not match, stop

A red **STOP — DO NOT POUR** means the cartridge is not from your run's lot. Set it aside, tell your lead, and press *Wrong cartridge — pulled it, nothing poured*. That records the catch. If your lead decides the pour goes ahead, choose the reason, add the detail and photograph the stamp — the log saves with the flag attached.

7 Enter your counts and submit

Good units filled, plus scrap empty and scrap filled kept separate — they mean different things to quality. Add anything worth noting, then submit. Your totals and the run's progress both move immediately.

The rest of your terminal

WHEN THE LINE STOPS

Log the downtime

Reason, minutes, and anything useful. Log it as it happens rather than reconstructing it later: downtime you do not record shows up as poor yield instead of a machine problem, and the ranking management uses to prioritise fixes is built from these entries.

CHANGEOVERS AND SPILLS

Photo audits

Start of shift, end of shift, moving a line onto a different resin, or a spill. These are the record that a changeover was done cleanly.

QUESTIONS

Manager comms

A direct thread with the plant lead inside the terminal, so the answer stays attached to your account rather than living in a text message.

YOUR NUMBERS

Today's performance

Units poured, scrap and quality yield at the top of your screen, updating as you log.

WORTH KNOWING

Your logs are attached to you, not to a spreadsheet cell someone can overwrite.

RPS bulk jugs skip the lot check — they carry no lot label, so the section does not appear.

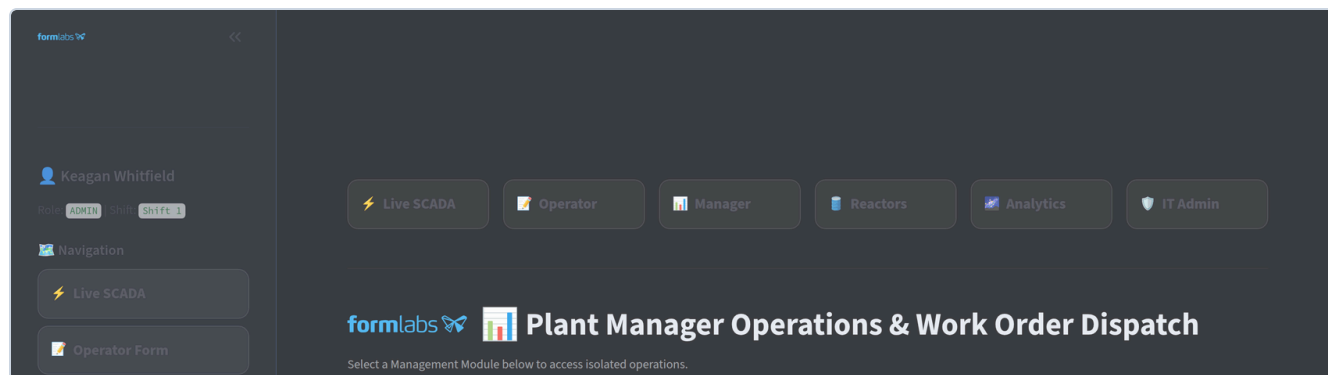
Nothing you submit is silently discarded. If a log cannot be matched to an active run you are told on screen, rather than finding out later that a progress bar never moved.

THE ONE RULE

Log it when it happens. An hour reconstructed at the end of a shift is a guess, and every pace figure, yield number and downtime decision downstream is built on what you enter.

Dispatching the day

Operator Pours resin at a pump station	Manager Dispatches and reviews	Admin Runs the system
--	--	---------------------------------



Manager Cockpit. Each management module on its own screen rather than one page trying to do everything.

1

Create the work orders

Reactor and size, resin, container format, target units, pump station and lot number. Assign an operator for accountability, but note progress follows the *station* — anyone working that pump contributes to the same run.

2

Enter the lot exactly as stamped

The most important field on the form: it is what the operator's cartridge check is compared against. Enter the number as printed after **L-** on the base. Spacing, case and the prefix are normalised, but a wrong number cannot be.

3

Watch the fleet, not the logs

The work order page rolls up fleet progress, active and queued counts, tracked pumps and operators ready. Drill into a run only when its progress stalls.

IF A PROGRESS BAR IS NOT MOVING

Almost always one of three things: the operator's station does not match the run's, the resin or container format differs, or the lot on the run does not match what is staged. The verification review shows which.

Reading the floor, closing the day

DAILY

Flagged lot checks

Mismatches poured anyway and cartridges pulled, each with the photograph, both lot numbers and the operator's reason. A mismatch that is really a formatting difference means the run's lot needs correcting, not the operator.

DAILY

Scrap and downtime

Yield against target, the empty-versus-filled split, and downtime by reason. Empty scrap is a handling problem; filled scrap is lost resin and considerably more expensive.

WEEKLY

Verification coverage

Full checks versus confirmation taps per operator. A station showing almost no full checks is one taking the fast path every hour.

WEEKLY

Trends

Output trend, share by resin, and per-operator history. The historical page covers anything beyond the current fortnight.

4

Close out finished runs

Complete a run on its real final count when the physical number differs. The correction is booked as a labelled system adjustment and the reactor is released.

5

Send the handover

Generate the PDF and dispatch it to the recipient list. Use the print-friendly layout for anything read on paper or forwarded outside the plant.

6

Answer the floor

The communications page holds every operator's thread. A question left unanswered there gets solved badly on the floor instead.

Keeping it running

Operator

Pours resin at a pump station

Manager

Dispatches and reviews

Admin

Runs the system

ON JOINING

Provision the account

Create the user with the correct role and shift before their first shift. Role decides which screens exist for them; shift decides which checklist and pace target apply.

ON LEAVING

Remove access, keep history

Deleting a user removes access. Production history stays attached through the foreign key on every log.

WEEKLY

Verify a backup

Take a snapshot and confirm it lands. A backup that has never been restored is a hypothesis; the restore path is on the same page.

PER RELEASE

Let migrations run

Schema changes apply on start-up. After deploying, start the app once and confirm it comes up before the shift arrives.

SEASONALLY

Review plant parameters

Shift times, lengths, break minutes, rate and yield targets. Every pace indicator and yield judgement is measured against these.

MONTHLY

Clear the suggestions inbox

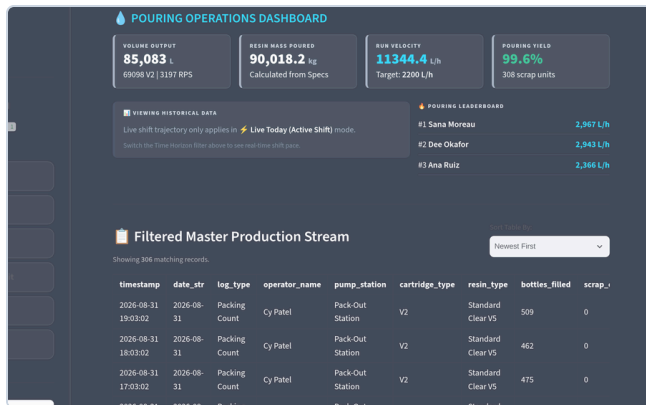
Staff file requests, bugs and floor issues from inside the app. An inbox nobody empties is one nobody writes to.

NOT THIS

Do not edit production rows directly in the database to fix a number. Use log management to remove the bad entry — it re-syncs the affected work order — or a reconciliation to book a variance. Both leave a visible record; a direct edit leaves the totals right and the story wrong.

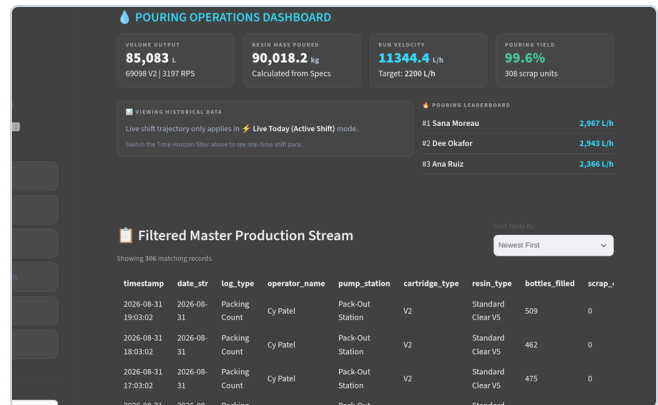
Twenty-four themes

Selectable per user and remembered across sessions. This is less cosmetic than it sounds: a station tablet under bright shop lighting, an office monitor and a wall display twenty feet from the viewer are three different readability problems. **Formlabs Forge** is the house theme and the one this document is set in.



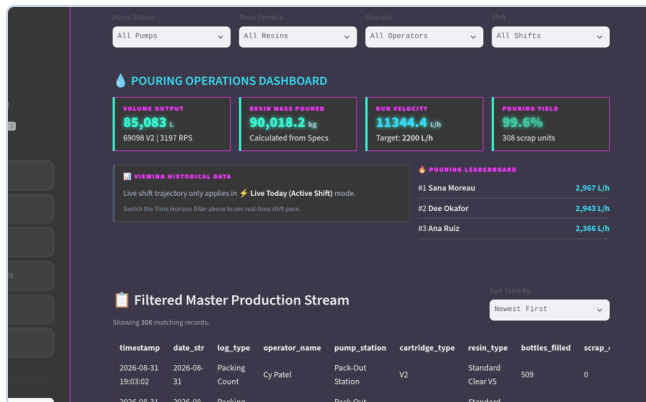
Formlabs Forge

#0F172A



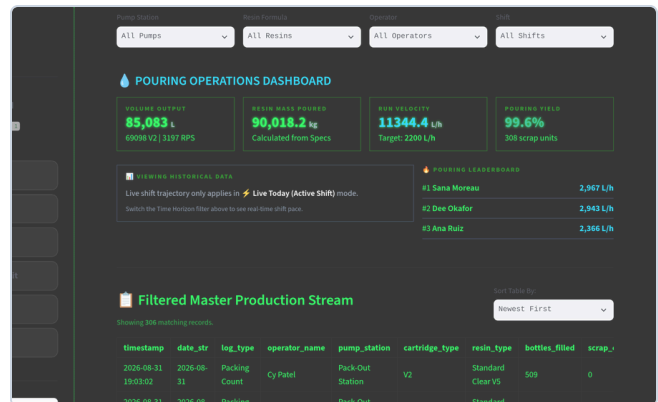
Formlabs Onyx

#0B0B0C



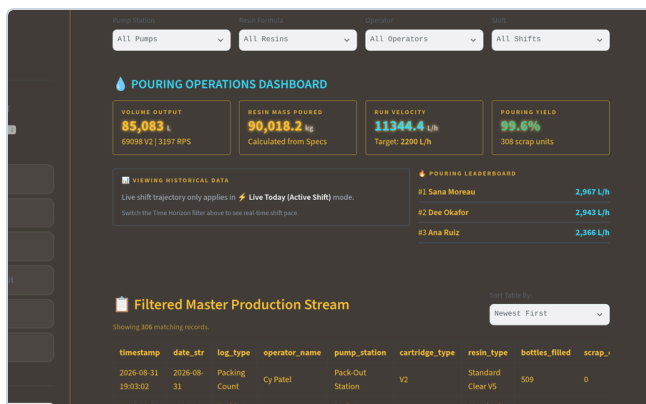
Neon Cyberpunk

#090117



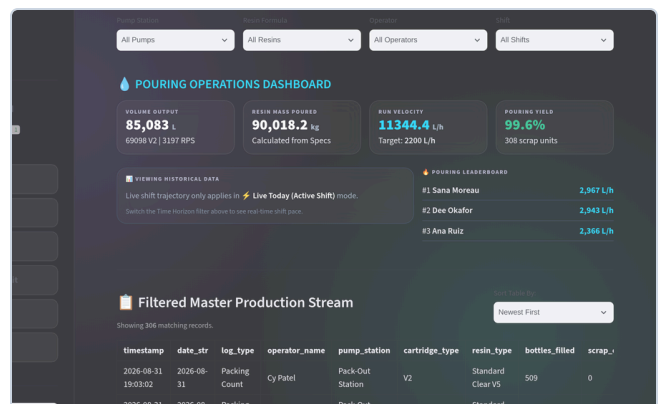
The Matrix

#000000



Amber CRT

#0D0800



Glass Aurora

gradient



Six of twenty-four shown, all rendering the same Live SCADA screen. The remainder include Deep Space, Titanium Alloy, Carbon Fiber, Resin Flow, Powder Bed, Ocean Depths, Dracula, Sapphire Enterprise, Emerald Ledger and Obsidian Chrome.

How it is tested

Before a release the system is exercised end to end against a throwaway database built from nothing each time. It refuses to run against the production connection.

75

workflow assertions — migrations, a simulated shift, every figure recomputed from raw rows

31

interface assertions driving the real operator screen headlessly

18

screens rendered and checked for errors on every pass

What the simulation covers

All four migrations are applied to an empty database and the result compared table by table against the application's own model definitions. A manager then creates three work orders; operators clear per-station checklists and pour — clean logs, a fast-path log, a mismatched cartridge, a cartridge pulled before pouring, a log with no run behind it, and an RPS pour that skips the check — alongside downtime. Every figure the interface displays is then recomputed from the raw rows and compared against hand-figured expectations: poured, scrap, yield, per-operator totals, run progress, flag rate and variance.

What that has caught

- **The run card was printing the lot the check was hiding.** An operator could have read the expected lot off the screen instead of the cartridge, which would have made the whole check decorative. Found because the interface test asserts the lot appears nowhere on the page.
- **Plant settings could never be saved.** The Admin Panel's save button passed a dictionary to a function expecting fifteen positional arguments, so shift times, rate targets and yield targets had never been changeable from the interface.
- **A content-delivery fetch could take the terminal down.** An unguarded network call for a decorative animation ran on every interaction; a floor PC that lost its internet would have hung the terminal before a control rendered.
- **Eight pages failed on a theme change,** and log cleanup crashed whenever a filter matched fewer than ten records — both long-standing.

KNOWN GAP

The page sweep renders every screen but does not submit forms, which is how the settings save stayed broken. Driving the main forms through submission is the next thing to close.

Experiments, not commitments

Two things exist in the codebase that are **not in use and not decided on**. They are described here so the capability is known, not because there is a plan attached to either. Both were built to find out whether the approach works.

Pack-out logging

A packing role, a pack-out logging tab, and skid conversion using each resin's units-per-skid figure. Poured and packed totals are counted separately, and the gap between them reports as floor work-in-progress. A reconciliation flow books the variance between what was poured for a lot and the final packed count as a single labelled system adjustment.

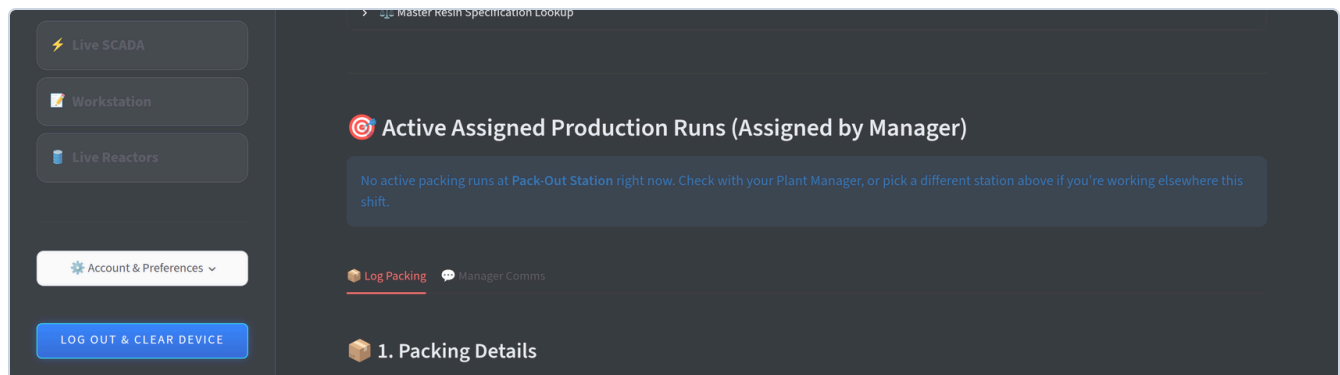
It is switched off. Turning it on is a single plant setting, but nobody has decided whether pack-out should be logged this way, and the workflow has not been tested against how the line actually runs.

Machine integration

A protocol-agnostic gateway for reading floor equipment directly, so counts and weights would not have to be typed. Each machine is a row in a registry and the polling logic sits behind one shared interface, with adapters written for Modbus TCP, Modbus RTU, OPC-UA, MQTT, Serial ASCII and HTTP. Readings would write through the same logging path a tablet uses, so analytics and run tracking would need no changes.

None of it has been tested against real equipment.

The adapters were written to cover whatever the scales or pump controllers turn out to speak; which of them is correct is unknown until someone connects one.



Pack-out logging, currently disabled. Container format, resin, lot and units packed, with the skid equivalent calculated as the count is typed.

READ THIS AS

Neither of these is a roadmap item or a commitment of dates. They are experiments that happen to be finished enough to demonstrate, and either could be abandoned without affecting anything in service.